

CURRICULUM REVISION 2021 - PROGRAMME SCHEME

STATE BOARD OF TECHNICAL EDUCATION, KERALA

TEACHING AND EXAMINATION SCHEME FOR DIPLOMA PROGRAMME

PROGRAM NAME : Electrical & Electronics Engineering

PROGRAM CODE : EE

PATTERN: FULL TIME

DURATION OF PROGRAMME: 3 YEARS (6 SEMESTERS)

SEMESTER: First Semester

SCHEME: Revision 2021

Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Humanities & Social Sciences	1001	Communication Skills in English	4	0	0	4	50	75	125	T	B	4
2.	Basic Science	1002	Mathematics I	5	0	0	5	50	75	125	T	B	5
3.	Basic Science	1003	Applied Physics I	3	0	0	3	50	75	125	T	B	3
4.	Basic Science	1004	Applied Chemistry	3	0	0	3	50	75	125	T	B	3
5.	Engineering Science	1005	Engineering Graphics	0	0	3	3	50	75	125	D	B	1.5
6.	Basic Science	2006	Applied Physics Lab	0	0	2	2	^	^	^	P	I	^
7.	Basic Science	1007	Applied Chemistry Lab	0	0	2	2	75	50	125	P	I	1
8.	Engineering Science	1008	Introduction to IT systems Lab	1	0	2	3	75	50	125	P	I	2
9.	Engineering Science	2009	Engineering Workshop Practice	0	0	3	3	^	^	^	P	I	^
10.	Humanities & Social Sciences	1009	Sports and Yoga	0	0	2	2	75	50	125	P	I	1
				16	0	14	30						20.5

PROGRAM NAME : Electrical & Electronics Engineering										PROGRAM CODE : EE			
SEMESTER: Second Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Basic Science	2002	Mathematics II	4	0	0	4	50	75	125	T	B	4
2.	Basic Science	2003	Applied Physics II	3	0	0	3	50	75	125	T	B	3
3.	Common Courses	2001	Environmental Science	3	0	0	3	50	75	125	T	I	0
4.	Engineering Science	2031	Fundamentals of Electrical & Electronics Engineering	3	0	0	3	50	75	125	T	B	3
5.	Engineering Science	2032	Elementary Concepts of Electrical Systems	3	0	0	3	50	75	125	T	B	3
6.	Humanities & Social Sciences	2008	Communication Skills in English Lab	0	0	3	3	75	50	125	P	I	1.5
7.	Basic Science	2006	Applied Physics Lab	0	0	2	2	75	50	125	P	I	1
8.	Engineering Science	2039	Fundamentals of Electrical & Electronics Engineering Lab	0	0	3	3	75	50	125	P	I	0
9.	Engineering Science	2038	Engineering Graphics using CAD software	0	0	3	3	75	50	125	P	I	0
10.	Engineering Science	2009	Engineering Workshop Practice	0	0	3	3	75	50	125	P	I	1.5
				16	0	14	30						17
Summer Internship I to be conducted after Semester 2													
1.	Common Courses	3009	Internship I	3 Weeks				0	75	75	P	I	2

PROGRAM NAME : Electrical & Electronics Engineering										PROGRAM CODE : EE			
SEMESTER: Third Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
L	T	P	Total										
1.	Programme core course	3031	Analog & Digital Circuits	3	1	0	4	50	75	125	T	B	3
2.	Programme core course	3032	DC Machines & Traction Motors	2	1	0	3	50	75	125	T	B	3
3.	Programme core course	3033	Fundamentals of Electric Circuits	3	1	0	4	50	75	125	T	B	4
4.	Programme core course	3034	Electrical & Electronics Measuring Instruments	3	1	0	4	50	75	125	T	B	4
5.	Programme core course	3035	Mechanical Engineering	2	1	0	3	50	75	125	T	I	0
6.	Programme core course	3036	DC Machines Lab	0	0	3	3	75	50	125	P	B	1.5
7.	Programme core course	3037	Electrical Measurements Lab	0	0	3	3	75	50	125	P	B	1.5
8.	Programme core course	3038	Electrical Workshop Practice	0	0	3	3	75	50	125	P	B	1.5
9.	Programme core course	3039	Mechanical Engineering Lab	0	0	3	3	75	50	125	P	B	1.5
				13	5	12	30						20

PROGRAM NAME : Electrical & Electronics Engineering										PROGRAM CODE : EE			
SEMESTER: Fourth Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
L	T	P	Total										
1.	Programme core course	4031	Power Electronics Devices and Circuits	3	1	0	4	50	75	125	T	B	4
2.	Programme core course	4032	Electrical Installation Design & Estimation	3	1	0	4	50	75	125	T	B	4
3.	Programme core course	4033	Induction Machines	3	1	0	4	50	75	125	T	B	4
4.	Common Courses	4001	Community Skills in Indian knowledge system	2	0	0	2	50	75	125	T	I	0
5.	Programme core course	4036	Electronics Lab	0	0	3	3	75	50	125	P	B	1.5
6.	Programme core course	4037	Induction Machines Lab	0	0	3	3	75	50	125	P	B	1.5
7.	Programme core course	4038	Domestic Appliances Repair & Maintenance Workshop	0	0	3	3	75	50	125	P	B	1.5
8.	Programme core course	4039	Professional Practice Lab	0	0	3	3	75	50	125	P	I	0
9.	Minor project	4009	Minor Project	0	0	4	4	0	75	75	P	I	2
				11	3	16	30						18.5
Summer Internship II to be conducted after Semester 4													
1.	Common Courses	5009	Internship II	4 Weeks				0	75	75	P	I	3

PROGRAM NAME : Electrical & Electronics Engineering										PROGRAM CODE : EE			
SEMESTER: Fifth Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Credits	
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		Evaluation Mode
L	T	P	Total										
1.	Common Courses	5001	Industrial Management and Safety	3	1	0	4	50	75	125	T	I	0
2.	Programme core course	5031	Synchronous Machines & FHP Motors	3	1	0	4	50	75	125	T	B	4
3.	Programme core course	5032	Electricity Generation, Transmission & Distribution	3	1	0	4	50	75	125	T	B	4
4.	Program Elective course	5033A	Renewable Energy Power Plant	4	0	0	4	50	75	125	T	B	4
	Program Elective course	5033B	Industrial Drives & Control	4	0	0	4	50	75	125	T	B	4
	Program Elective course	5033C	Switch Gear & Protection	4	0	0	4	50	75	125	T	B	4
5.	Programme core course	5037	Synchronous Machines Lab	0	0	3	3	75	50	125	P	B	1.5
6.	Programme core course	5038	Industrial Electrical Engineering Lab	0	0	3	3	75	50	125	P	B	1.5
7.	Program Elective course	5039A	Electrical Estimation Lab	0	0	3	3	75	50	125	P	B	1.5
	Program Elective course	5039B	Applied Electronics Lab	0	0	3	3	75	50	125	P	B	1.5
	Program Elective course	5039C	Solar Energy Technology Lab	0	0	3	3	75	50	125	P	B	1.5
8.	Seminar	5008	Seminar	0	0	2	2	0	75	75	P	I	1
9.	Major Project	6009	Major Project	0	0	3	3	0	0	0	P	I	0
				13	3	14	30						17.5

The Student has two options to complete the sixth semester and he/she shall select any one: 1. NORMAL PATHWAY 2. INTERNSHIP PATHWAY

1. NORMAL PATHWAY

PROGRAM NAME : Electrical & Electronics Engineering

PROGRAM CODE : EE

SEMESTER: Sixth Semester

SCHEME: Revision
2021

Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Humanities & Social Sciences	6001	Entrepreneurship and Startup	3	1	0	4	50	75	125	T	I	4
2.	Program Elective course	6031A	Energy Conservation & Audit (EE)	3	1	0	4	50	75	125	T	B	4
	Program Elective course	6031C	Microcontroller & PLC	3	1	0	4	50	75	125	T	B	4
	Program Elective course	6031D	Electric Vehicles	3	1	0	4	50	75	125	T	B	4
3.	Open Elective course	6032A	Solar Power Technologies	4	0	0	4	50	75	125	T	I	4
	Open Elective course	6032B	Energy Conservation & Management	4	0	0	4	50	75	125	T	I	4
	Open Elective course	6032C	Electrification of Residential Buildings	4	0	0	4	50	75	125	T	I	4
	Open Elective course	6032D	Electric Vehicles & Traction	4	0	0	4	50	75	125	T	I	4
4.	Common Courses	6002	Indian Constitution	2	0	0	2	50	75	125	T	I	0
5.	Programme core course	6037	Electrical Computer Aided Drafting Lab	0	0	3	3	75	50	125	P	I	1.5
6.	Programme core course	6038	Industrial Automation Lab	0	1	3	4	75	50	125	P	I	2.5
7.	Program Elective course	6039A	Applied Electrical Testing Lab	0	0	3	3	75	50	125	P	B	1.5
	Program Elective course	6039B	Modelling and simulation Lab	0	0	3	3	75	50	125	P	B	1.5
	Program Elective course	6039C	Advanced Solar Photovoltaic Lab	0	0	3	3	75	50	125	P	B	1.5
8.	Major Project	6009	Major Project	0	0	6	6	75	50	125	P	I	4
				12	3	15	30						21.5

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2. INTERNSHIP PATHWAY													
PROGRAM NAME : Electrical & Electronics Engineering											PROGRAM CODE : EE		
SEMESTER: Sixth Semester											SCHEME: Revision 2021		
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Humanities & Social Sciences	6001	Entrepreneurship and Startup	3	1	0	4	50	75	125	T	I	4
2.	Common Courses	6002	Indian Constitution	2	0	0	2	50	75	125	T	I	0
3.	Programme core course	6037	Electrical Computer Aided Drafting Lab	0	0	3	3	75	50	125	P	I	1.5
4.	Programme core course	6038	Industrial Automation Lab	0	1	3	4	75	50	125	P	I	2.5
5.	Internship	6007	Internship	4 - 6 months per semester				250	250	500	P	I	13.5
				5	2	6	13						21.5
Teaching Scheme : L-Lecture, T-Tutorial, P- Practical Examination Type : T-Theory, P-Practical, D-Drawing Examination Mode : B-Board Level, I-Institution Level ^ Carried over to next semester													